

René Gaudoin, PhD

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| London

| [LinkedIn](#)

| [GitHub](#)

Personal Profile

- **Objective:** Building innovative solutions that push technical boundaries — exploring healthcare, science, fine wine, and beyond.
- **Strength:** Finding and implementing novel solutions based on diverse experience in academia, analytics consulting and startups; driven by curiosity.
- **Academic background:** Research in computational science (solid state physics, quantum chemistry, public health, theoretical neurobiology); publications in high-impact journals (e.g. PRL); university teaching and tutoring.
- **Non-academic background:** DS/ML at QuantumBlack/McKinsey; online advertising; CAIO at StratosX (airline disruption recovery); currently Staff DS at Onsera (GLP-1 patient programmes); mentoring junior colleagues.
- **Wine expertise:** 100% in [WSET 3](#); won an Oddbins Scholarship prize in the follow-up round for top scorers (2016).
- **Highlights:** Deep Reinforcement Learning ([America's Cup 2021](#), DataIQ Grand Prix), survival analysis, causal modelling and real world evidence on longitudinal medical data, deep neural network analysis of 1d medical data, randomised control trial simulation.
- **Languages:** German/English bilingual; Spanish fluent; French intermediate.

Professional Experience and Achievements

- **Machine learning and statistical methods:** Reinforcement learning (supply chain optimisation, logistics, sailing); health care (public health, HES, real world evidence, randomised control trial simulations, drug discovery, marketing), LLMs (LangChain, agentic workflows, prompt engineering); deep learning & DL surrogates, supervised/unsupervised tech stack (including causal modelling/inference; Bayesian stats etc.).
- **Developing new algorithms:** Calibration of statistical data, exact sampling for Quantum Monte Carlo, extending the Perceptron algorithm to time series of realistic neurons, modelling with aggregated target variable, and more.
- **Computational science:** Ab initio methods (DFT, post-HF, QMC) and their theoretical foundations, theoretical neurobiology, basic computational fluid dynamics.
- **Tech stack:** *Languages:* Python, C, Fortran, Scala, some R, SQL. *Deep learning:* TensorFlow/Keras, PyTorch. *ML/Data:* Ray/RLlib, Stable-Baselines, pandas, scikit-learn, scipy, statsmodels, seaborn, matplotlib. *Infrastructure:* Git, Linux, Jupyter, AWS, Databricks, Spark, L^AT_EX.

Employment and Work Experience

Staff Data Scientist, Onsera, London

4.2026–present

Health-tech startup helping patients make the best use of GLP-1 treatment through a structured programme.

- Robust patient health scores targeting one or more outcomes, combinable with established indices (e.g. Elixhauser comorbidity) and stable under variable per-patient look-back
- Forward-looking adherence models with flexible, calibrated predictive distributions
- Causal, survival and autoregressive (time-series) modelling on longitudinal patient data for side effects, mortality and other emerging clinical and product questions

CAIO, StratosX, London

2.2025–1.2026

Chief AI Officer at airline disruption recovery startup. Defining, implementing, and validating ML applications across the platform.

Independent research, London/Munich

10.2024–2.2025

Time out for family; independent exploration of ideas and emerging technologies.

Data Scientist, QuantumBlack (McKinsey), London 5.2017–9.2024

End-to-end ML solutions from ideation to production code handover.

Selected projects:

- America's Cup 2021 (Team NZ): deep RL for boat design — DataIQ Grand Prix winner
- Major European pharma corporation: marketing optimisation with novel algorithm for mis-aligned units of analysis — assetised internally
- Major pharma corporation: real-world evidence study with survival modelling for heart failure
- Major UK university (R&D collaboration): ML/deep learning analysis of clinical haematology sensor data
- Major metals company: deep RL for order processing
- European generics manufacturer: market share optimisation

Other areas: LLM prototyping (LangChain agents, exploring RL for autonomous web navigation), RCT simulation, IT productivity, DL surrogates, CFD.

Data Scientist, Infectious Media, London 9.2014–5.2017

Modelling click/conversion probabilities for bidding on online ads (Python/Scala/Spark)

Fellow, S2DS data science summer school, London 8.2014

Business, IT, and ML lectures followed by project work for Stratified Medical (Gensim, Neo4j).

Post-Doc (medical statistics), Imperial College (Dr. Foster Unit), London 2.2012–3.2014

ML for risk adjustment of standardised mortality ratios on national hospital data (HES).

Post-Doc, University of Vienna (Computational Material Physics) 5.2010–1.2012

DFT and post-Hartree-Fock methods extending the VASP package.

Academic Visitor, Fritz-Haber-Institut, Berlin 9.2009–3.2010

Research into post-HF methods for van-der-Waals dominated systems.

Post-Doc, Donostia International Physics Institute 11.2004–3.2009

New algorithms for parallelised Quantum Monte Carlo (CASINO package), including a new exact sampling technique.

Post-Doc, Rutgers University, NJ, USA 12.2002–9.2004

Fundamentals of density functional theory: showed that the Hohenberg–Kohn theorem does not generalise to excited states (PRL 2004).

Research Assistant, Imperial College, London 1.2001–3.2002

Publishing research and finalising papers at the Condensed Matter Theory Group.

Independent Research

Independent research, London 2.2025–present

Ongoing alongside employment (selected code on [GitHub](#)): exploration in sparse-reward reinforcement learning, extensions of Monte Carlo tree search, game equilibria via differentiable game trees, time-series and survival modelling for patient embeddings and generative health-data models.

Education

PhD, Imperial College, London 10.1996–12.2000

Condensed Matter Theory Group – Quantum Monte Carlo calculations. We developed an inhomogeneous extension to trial wavefunctions implemented in the Casino package.

Diplom-Physiker, TU München 12.1988–2.1995

General Physics (Maths, Physics, some Chemistry), grade 1.6. Final year thesis in theoretical Biophysics modelling spiking neurons and developing a Perceptron for these.

Secondary school, European School 10.1980–6.1988

European Baccalaureate (top marks in economics, maths, physics, and chemistry)